Information Geometry of the Quantum-Classical Boundary in Biological Information Processing

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Abstract

Decoherence drives the quantum-to-classical transition, but its information-geometric structure—the precise dimensional and metric changes it imposes on quantum state space—has not been quantified in the Bures/Fisher–Rao framework. We prove that decoherence is a dimensional collapse: the dephasing channel maps a (d^2-1)-dimensional quantum manifold onto the $(d-1)$-dimensional classical probability simplex, annihilating d^2-d coherence directions (*proved*: Theorems 3.1, 3.4, 3.6). We compute the exact Bures principal angle at every full-rank qubit state and show that for $d \geq 3$, Bures-orthogonality holds if and only if ρ is diagonal. Since coherences store free energy $k_B T \cdot C_r(\rho)$, decoherence has a thermodynamic cost that scales as $\sim \ln d/d^2$ per collapsed dimension (*proved*: §4). Applying this framework to validated photosynthetic Hamiltonians—the Fenna–Matthews–Olson complex ($d = 7$, collapse $48 \to 6$) and the PE545 cryptophyte antenna ($d = 8$, collapse $63 \to 7$)—we find that at the ENAQT-optimal dephasing rate, all principal angles exceed 87° (FMO) and 89° (PE545): biologically optimal transport

occurs in a regime that is geometrically close to classical (*empirical*). Extending to cortical gamma oscillations ($d = 10$, collapse $99 \to 9$), biological dephasing places neural oscillations ∼12 orders of magnitude past the boundary at oscillation-frequency scales (*empirical*), but the molecular machinery they coordinate—ion channels, synaptic receptors—operates at energy scales where $\gamma_{\text{bio}}/J_{\max} \sim O(1\text{–}30)$ (*hypothesis*). To make this concrete, an illustrative ion-channel selectivity filter model ($d = 4$, Appendix C.1) gives $\theta_{\min} \approx 88.5°$ and $\chi \approx 2.8 \times 10^{-4}$ at biological dephasing (*model-based empirical*), i.e. non-classical ($\chi > 0$) but near-classical geometry. We define a non-classicality index χ from the principal angles and a coordinated dimensionality D_{coord} that scales with the number of entrained molecular sites N; the deep classicality of oscillations ($\chi_{\text{osc}} \approx 0$) is then a feature, providing a geometrically reliable channel linking molecular-scale non-classical dynamics to population-level computation. An illustrative chromatin molecular-oscillator case study (Appendix C) supports the energy-scale argument that molecular biology can sit near the quantum-classical boundary, though the geometric position is sensitive to coupling strength. The ENAQT regime may represent a cost-benefit sweet spot where extra dimensional capacity from partial coherence comes at modest thermodynamic cost; whether this reflects evolutionary optimization is testable (*hypothesis*).

1 Introduction

The quantum-to-classical transition mediated by decoherence [1, 2, 3, 4] is not merely a theoretical curiosity in biological systems—it is functionally relevant. Photosynthetic complexes exploit quantum coherence during energy transfer on femtosecond timescales [5], and optimal transport efficiency occurs at intermediate dephasing rates, a phenomenon termed environment-assisted quantum transport (ENAQT) [6, 7, 8]. The quantum biology literature has established that partial decoherence can enhance biological function [9], but the geometric structure of this transition—how many degrees of freedom are destroyed, how cleanly classical information separates from quantum information, and what the dimensional architecture of intermediate regimes looks like—has not been

quantified.

Information geometry provides the natural mathematical language for these questions. The quantum Fisher information matrix (QFIM) [10, 11] measures the dimensionality of distinguishable quantum states, and the Bures metric it induces [12, 13, 14] is the unique minimal monotone Riemannian metric on quantum state space. The classical Fisher–Rao metric [15, 16, 17] plays the analogous role for probability distributions. Despite the maturity of both decoherence theory and information geometry, these frameworks have not been combined to characterize the dimensional structure of the quantum-to-classical transition.

In this paper, we show that decoherence is a *dimensional collapse* in the information-geometric sense. A d-dimensional quantum system lives on a (d^2-1)-dimensional state manifold equipped with the Bures metric. Decoherence projects this manifold onto the $(d-1)$-dimensional classical probability simplex equipped with the Fisher–Rao metric. Specifically: *Collapse.* The decoherence channel reduces the QFIM rank from d^2-1 to at most $d-1$ (Theorem 3.2), and the Bures metric restricts to the Fisher–Rao metric on Δ_{d-1} (Theorem 3.3). At full-rank diagonal states, the differential of the dephasing map is the Bures-orthogonal projector onto the classical tangent subspace (Theorem 3.4).

Geometry of the split. We compute the exact Bures principal angle θ_B between the classical and quantum tangent subspaces at every full-rank qubit state (Theorem 3.5). For $d \geq 3$, we prove that Bures-orthogonality holds if and only if ρ is diagonal (Theorem 3.8), and a perturbative formula quantifies the departure (Theorem 3.10).

Application. We apply the framework to three biological systems spanning 12 orders of magnitude in dephasing: two photosynthetic complexes—the Fenna–Matthews–Olson (FMO) complex ($d = 7$, collapse $48 \to 6$) and the PE545 cryptophyte antenna ($d = 8$, collapse $63 \to 7$)—and a cortical gamma-oscillation assembly ($d = 10$, collapse $99 \to 9$). At the ENAQT-optimal dephasing rate both photosynthetic complexes are geometrically nearly classical (all angles $> 87°$ for FMO, $> 89°$ for PE545). The neural system sits $\sim$12 orders of magnitude past the boundary, geometrically indistinguishable from the classical simplex at oscillation-frequency scales. We additionally include two illustrative

molecular models—ion channel selectivity filter (Appendix C.1) and chromatin dynamics (Appendix C)—to test the molecular-scale hypothesis.

Claims in this paper are organized into three tiers. *Proved*: mathematical theorems with full proofs (Theorems 3.1–3.7, §4 thermodynamics). *Empirical*: results computed from validated or physically motivated Hamiltonians (FMO, PE545, neural models). *Hypothesis*: interpretations connecting proved and empirical results to biological function (cost-benefit interpretation, evolutionary tuning). We tag each tier explicitly throughout.

The paper is organized as follows. Section 2 establishes the information-geometric framework. Section 3 contains the core mathematical results. Section 4 derives the thermodynamic cost of dimensional collapse. Section 5 applies the framework to FMO and PE545. Section 6 extends to neural oscillations. Section 7 discusses implications, limitations, and the cost-benefit interpretation of the ENAQT regime. Appendix C presents an illustrative molecular-oscillator case study extending the analysis beyond photosynthesis.

2 Information-Geometric Framework

We establish the mathematical framework that makes "dimensional collapse" precise.

Definition 2.1 (Quantum state manifold)**.** The space of $d \times d$ density matrices,

$$\mathcal{M}_Q = \left\{\rho \in \mathbb{C}^{d\times d} : \rho \geq 0,\ \operatorname{Tr}\rho = 1\right\}, \tag{1}$$

is a real manifold of dimension $d^2 - 1$: a $d \times d$ Hermitian matrix has d^2 real parameters, the trace constraint removes one, leaving $d - 1$ populations and $d(d-1)$ coherences.

Definition 2.2 (Quantum Fisher information matrix)**.** Let $\rho(\theta)$ be a smooth parametric family of density matrices with $\theta \in \mathbb{R}^p$. The QFIM is the $p \times p$ matrix with entries

$$[F_Q]_{ij} = \operatorname{Tr}\left[\rho\,\frac{L_iL_j + L_jL_i}{2}\right], \tag{2}$$

where L_i is the symmetric logarithmic derivative (SLD) defined by $\partial_i\rho = (\rho L_i + L_i\rho)/2$. For full-rank ρ with eigenbasis $\rho = \sum_k p_k|k\rangle\langle k|$, the matrix elements are $(L_i)_{mn} =$

$2(\partial_i\rho)_{mn}/(p_m + p_n)$.

The QFIM defines the Bures–Helstrom metric $ds^2_{\text{Bures}} = \frac{1}{4}\sum_{ij}[F_Q]_{ij}\, d\theta^i\, d\theta^j$, the unique minimal monotone Riemannian metric on quantum state space [14, 11]. By Petz's monotonicity theorem, for any CPTP map Λ:

$$F_Q\Big(\Lambda[\rho(\theta)]\Big) \leq F_Q\Big(\rho(\theta)\Big) \tag{3}$$

in the Löwner order—quantum channels cannot increase statistical distinguishability.

Definition 2.3 (Effective quantum dimensionality)**.** Given a QFIM F_Q with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_p \geq 0$, the effective quantum dimensionality is the participation ratio

$$D^Q_{\text{eff}} = \frac{\left(\sum_i \lambda_i\right)^2}{\sum_i \lambda_i^2}, \tag{4}$$

ranging from 1 (single dominant direction) to p (all directions equally estimable). D^Q_{eff} is a coordinate-dependent diagnostic: the rank of F_Q is invariant, but the participation ratio depends on the parameterization. The main theorems of this paper depend only on rank, kernel structure, and metric properties.

3 Decoherence as Dimensional Projection

We now show that decoherence acts as a geometric projection, annihilating the $d^2 - d$ off-diagonal directions and collapsing quantum state space onto the $(d-1)$-dimensional classical probability simplex.

Definition 3.1 (Decoherence channel)**.** Fix an orthonormal pointer basis $\{|k\rangle\}_{k=1}^d$. The complete dephasing channel is

$$\Lambda_{\text{dec}}(\rho) = \sum_{k=1}^{d} |k\rangle\langle k|\, \rho\, |k\rangle\langle k|, \tag{5}$$

setting all off-diagonal elements to zero: $[\Lambda_{\text{dec}}(\rho)]_{kl} = \delta_{kl}\, \langle k|\rho|k\rangle$.

Theorem 3.2 (Dimensional collapse)**.** *Let $\mathcal{M}_Q$ be the quantum state manifold of a d-dimensional system* ($\dim \mathcal{M}_Q = d^2 - 1$)*. The decoherence channel Λ_{dec} maps $\mathcal{M}_Q$ onto the classical probability simplex Δ_{d-1}* ($\dim \Delta_{d-1} = d - 1$)*. Specifically:*

(i) *The image $\Lambda_{\text{dec}}(\mathcal{M}_Q) = \Delta_{d-1}$.*

(ii) *The $d^2 - d$ off-diagonal parameters become unestimable: for any smooth parameterization $\theta = (\theta_{\text{diag}}, \theta_{\text{off}})$, the QFIM of $\Lambda_{\text{dec}}[\rho(\theta)]$ satisfies*

$$[F_Q(\Lambda_{\text{dec}}[\rho])]_{\alpha\beta} = 0 \quad \textit{whenever } \alpha \in \theta_{\text{off}} \textit{ or } \beta \in \theta_{\text{off}}. \tag{6}$$

In particular, $\operatorname{rank} F_Q(\Lambda_{\text{dec}}[\rho]) \leq d - 1$.

(iii) *The dimensional collapse ratio is*

$$\frac{\dim \Delta_{d-1}}{\dim \mathcal{M}_Q} = \frac{d-1}{d^2-1} = \frac{1}{d+1}, \tag{7}$$

which vanishes as $d \to \infty$.

Proof. Part (i): Λ_{dec} sends ρ to $\operatorname{diag}(\langle k|\rho|k\rangle)$, a point in Δ_{d-1}; every point in Δ_{d-1} is the image of some diagonal state. Part (iii) is arithmetic.

For Part (ii): the decohered state depends only on θ_{diag}, so $\partial\rho_{\text{dec}}/\partial\theta^{\alpha}_{\text{off}} = 0$, giving $L_\alpha = 0$ and hence $[F_Q]_{\alpha\beta} = 0$ whenever α or β is off-diagonal. The surviving diagonal block reduces to the classical Fisher information:

$$[F_Q]^{\text{diag}}_{ij} = \sum_{k=1}^{d} \frac{1}{p_k} \frac{\partial p_k}{\partial \theta^i_{\text{diag}}} \frac{\partial p_k}{\partial \theta^j_{\text{diag}}} = [F_{\text{cl}}]_{ij}, \tag{8}$$

which has rank at most $d - 1$. □

Theorem 3.3 (Metric restriction)**.** *Under Λ_{dec}, the Bures metric on $\mathcal{M}_Q$ restricts to the Fisher–Rao metric on Δ_{d-1}:*

$$ds^2_{\text{Bures}}\Big|_{\Delta_{d-1}} = \frac{1}{4} \sum_{k=1}^{d} \frac{dp_k^2}{p_k} = \frac{1}{4}\, ds^2_{\text{Fisher-Rao}}. \tag{9}$$

Proof. For diagonal states, the SLD is diagonal: $\langle k|L_i|k\rangle = (\partial_i p_k)/p_k$. The Bures line element becomes $\frac{1}{4}\sum_k dp_k^2/p_k$, the Fisher–Rao metric [11]. □

Theorem 3.4 (Orthogonal decomposition)**.** *Let $\rho = \mathrm{diag}(p_1, \ldots, p_d)$ be a full-rank diagonal state. The tangent space admits an orthogonal decomposition under the Bures metric:*

$$T_\rho \mathcal{M}_Q = T_\rho^{\mathrm{diag}} \oplus T_\rho^{\mathrm{off}}, \tag{10}$$

where T_ρ^{diag} is the $(d-1)$-dimensional traceless diagonal subspace and T_ρ^{off} is the (d^2-d)-dimensional off-diagonal Hermitian subspace. The differential $d\Lambda_{\mathrm{dec}}|_\rho$ acts as the identity on T_ρ^{diag} and as zero on T_ρ^{off}, so it is the g_B-orthogonal projector onto T_ρ^{diag}.

Proof. At a diagonal state, any traceless Hermitian X decomposes as $X = X_D + X_O$ (diagonal + off-diagonal). The SLD for a diagonal perturbation X_D is diagonal: $(L_{X_D})_{kk} = x_k/p_k$. The SLD for an off-diagonal perturbation X_O is off-diagonal: $(L_{X_O})_{mn} = 2z_{mn}/(p_m + p_n)$ for $m \neq n$. The anticommutator $\{L_{X_D}, L_{X_O}\}$ is off-diagonal (product of diagonal and off-diagonal matrices), so $\rho\{L_{X_D}, L_{X_O}\}$ is off-diagonal with zero trace:

$$g_B(X_D, X_O) = \frac{1}{4}\,\mathrm{Tr}\left[\rho\,\frac{\{L_{X_D}, L_{X_O}\}}{2}\right] = 0.$$

The dephasing channel preserves diagonal and annihilates off-diagonal perturbations, completing the identification as orthogonal projector. □

3.1 The classical–quantum split away from the simplex

Theorem 3.4 establishes orthogonality at diagonal states. What happens at states carrying coherence? For qubits, a single principal angle characterizes the full relationship.

Theorem 3.5 (Bures separation angle)**.** *Let $\rho = (I + \mathbf{r}\cdot\boldsymbol{\sigma})/2$ be a full-rank qubit state ($|\mathbf{r}| < 1$) with coherence magnitude $r_\perp = \sqrt{r_x^2 + r_y^2}$ and population imbalance r_z. The principal angle θ_B between T_ρ^{diag} and T_ρ^{off} under the Bures metric satisfies*

$$\cos^2\theta_B = \frac{r_z^2\, r_\perp^2}{(1 - r_\perp^2)(1 - r_z^2)}. \tag{11}$$

Equivalently,

$$\sin^2\theta_B = \frac{1-|\mathbf{r}|^2}{(1-r_\perp^2)(1-r_z^2)}. \tag{12}$$

Proof. The Bures metric on qubit state space in Bloch coordinates is $g^B_{ij} = \frac{1}{4}[\delta_{ij} + r_i r_j/(1-|\mathbf{r}|^2)]$. The principal angle is found by maximizing $g_B(\hat{e}_z, v)/(\|\hat{e}_z\| \cdot \|v\|)$ over $v \in T_\rho^{\text{off}}$. The maximizer is $v_\perp = (r_x, r_y, 0)/r_\perp$. Computing:

$$\begin{aligned} g_B(\hat{e}_z, v_\perp) &= \frac{r_z r_\perp}{4(1-|\mathbf{r}|^2)}, \\ g_B(\hat{e}_z, \hat{e}_z) &= \frac{1-r_\perp^2}{4(1-|\mathbf{r}|^2)}, \\ g_B(v_\perp, v_\perp) &= \frac{1-r_z^2}{4(1-|\mathbf{r}|^2)}. \end{aligned}$$

The squared cosine follows; the sine formula uses $(1-r_\perp^2)(1-r_z^2) - r_z^2 r_\perp^2 = 1-|\mathbf{r}|^2$. □

Corollary 3.6 (Structure of the classical–quantum split)**.**

(i) $\theta_B = \pi/2$ *if and only if* $r_\perp = 0$ *or* $r_z = 0$*: subspaces are Bures-orthogonal at all diagonal and all equatorial states.*

(ii) $\theta_B < \pi/2$ *if and only if both* $r_\perp \neq 0$ *and* $r_z \neq 0$.

(iii) *Dephasing monotonically increases* θ_B *toward* $\pi/2$.

Proof. Parts (i)–(ii) are immediate from (11). For (iii): under dephasing, $r_\perp \mapsto (1-2\gamma)r_\perp$ while r_z is preserved. The numerator of $\cos^2\theta_B$ decreases monotonically and the denominator increases, so θ_B increases. □

Remark 3.7 (Estimation-theoretic interpretation). For a full-rank qubit, eliminating coherence parameters via the Schur complement gives effective Fisher information $[F_Q]^{\text{eff}}_{zz} = [F_Q]_{zz}\sin^2\theta_B = 1/(1-r_z^2)$ [18, 19]: the Bures separation angle determines the fraction of population Fisher information surviving marginalization over coherence.

3.2 Generalization to qudits

For d-dimensional systems, there are $d-1$ principal angles between T_ρ^{diag} and T_ρ^{off}. Theorem 3.4 shows these are all $\pi/2$ at diagonal states. For $d \geq 3$, the converse holds

exactly:

Theorem 3.8 (Global orthogonality criterion for $d \geq 3$)**.** *For $d \geq 3$, a full-rank state ρ achieves the Bures-orthogonal decomposition $T_\rho^{\mathrm{diag}} \perp_{g_B} T_\rho^{\mathrm{off}}$ if and only if ρ is diagonal in the pointer basis.*

Proof. The reverse direction is Theorem 3.4. For the forward direction, assume orthogonality holds. Define the Jordan product map $\mathcal{J}_\rho(L) = (\rho L + L\rho)/2$, which is invertible on Hermitian matrices when ρ is full-rank. The SLD for tangent vector X is $L_X = \mathcal{J}_\rho^{-1}(X)$.

For any traceless diagonal $X_D = \mathrm{diag}(x_1, \ldots, x_d)$ and off-diagonal Hermitian X_O, the SLD metric identity gives

$$g_B(X_D, X_O) = \tfrac{1}{2} \sum_k x_k \, (L_{X_O})_{kk}. \tag{13}$$

Since this vanishes for all traceless x, orthogonality requires $(L_{X_O})_{kk}$ to be constant across k—the diagonal of L_{X_O} is flat. Hence $L_{X_O} = cI + Z$ for some $c \in \mathbb{R}$ and Z off-diagonal.

Let $\mathcal{O}$ denote the $d(d-1)$-dimensional space of off-diagonal Hermitian matrices and $\mathcal{W} = \mathbb{R}I \oplus \mathcal{O}$. The flat-diagonal condition means $S := \mathcal{J}_\rho^{-1}(\mathcal{O}) \subset \mathcal{W}$. Consider the projection $\pi \colon \mathcal{W} \to \mathcal{O}$ defined by $cI + Z \mapsto Z$. This is injective on S: if $cI \in S$ then $\mathcal{J}_\rho(cI) = c\rho \in \mathcal{O}$, but ρ has positive diagonal entries, forcing $c = 0$. Since $\dim S = \dim \mathcal{O} = d(d-1)$, the restricted projection $\pi|_S$ is an isomorphism.

Therefore, for every $Z \in \mathcal{O}$ there exists a unique $c(Z) \in \mathbb{R}$ with $c(Z)I + Z \in S$, meaning $\mathcal{J}_\rho(c(Z)I + Z) \in \mathcal{O}$. Taking the diagonal:

$$\mathrm{diag}(\rho Z + Z\rho) = -2c(Z)\, p, \qquad p = (p_1, \ldots, p_d) = \mathrm{diag}(\rho). \tag{14}$$

Now substitute the basis elements. For $R_{kl} = |k\rangle\langle l| + |l\rangle\langle k|$:

$$\mathrm{diag}(\rho R_{kl} + R_{kl}\rho) = 2\,\mathrm{Re}(\rho_{kl})\,(e_k + e_l), \tag{15}$$

and for $I_{kl} = i|k\rangle\langle l| - i|l\rangle\langle k|$:

$$\mathrm{diag}(\rho I_{kl} + I_{kl}\rho) = 2\,\mathrm{Im}(\rho_{kl})\,(e_k + e_l). \tag{16}$$

If $\rho_{kl} \neq 0$, at least one of these is nonzero, so $e_k + e_l \in \mathrm{span}\{p\}$. But for $d \geq 3$, the vector $e_k + e_l$ has $d - 2 \geq 1$ zero entries while p has all entries strictly positive, so $e_k + e_l$ cannot be proportional to p. Contradiction; hence $\rho_{kl} = 0$ for all $k \neq l$. □

Remark 3.9 (The qubit exception). For $d = 2$, the vector $e_1 + e_2 = (1, 1)$ is proportional to $p = (p_1, p_2)$ if and only if $p_1 = p_2$, recovering the qubit result: equatorial states ($r_z = 0$) achieve Bures-orthogonality despite carrying coherence (Corollary 3.6(i)). The proof above shows that $d \geq 3$ eliminates this loophole.

Theorem 3.8 settles the qualitative question: for $d \geq 3$, any coherence breaks orthogonality. We now quantify *how much* the subspaces mix.

Theorem 3.10 (Perturbative cross-Gram near the simplex)**.** *Let $\rho = \mathrm{diag}(p) + C$ be a full-rank d-dimensional state with off-diagonal part C.*

(i) *Near the simplex ($\|C\|$ small), the cross-Gram matrix has entries*

$$g_B(D_a, R_{kl}) = -\frac{\mathrm{Re}[C_{kl}]}{2(p_k + p_l)}\left(\frac{x_a^k}{p_k} + \frac{x_a^l}{p_l}\right) + O(\|C\|^2), \tag{17}$$

where $D_a = \mathrm{diag}(x_a^1, \ldots, x_a^d)$ is a traceless diagonal tangent vector and $R_{kl} = |k\rangle\langle l| + |l\rangle\langle k|$.

(ii) *Under partial dephasing $C_{kl} \mapsto (1-2\gamma)C_{kl}$, the cross-Gram entries scale as $(1-2\gamma)$ at leading order, and all principal angles converge to $\pi/2$ as $\gamma \to 1/2$.*

Proof. Part (i): Expand $L_a = L_a^{(0)} + L_a^{(1)} + O(\|C\|^2)$. At zeroth order, $(L_a^{(0)})_{kk} = x_a^k/p_k$ with zero off-diagonal entries. At first order, the Lyapunov equation gives $(L_a^{(1)})_{mn} = -C_{mn}(x_a^m/p_m + x_a^n/p_n)/(p_m + p_n)$ for $m \neq n$. For R_{kl}, the first-order diagonal corrections are $(L_R^{(1)})_{kk} = -2\mathrm{Re}[C_{kl}]/[p_k(p_k + p_l)]$. Summing the three first-order contributions to $g_B(D_a, R_{kl})$ yields (17).

Part (ii): Under $C \mapsto (1-2\gamma)C$, equation (17) scales as $(1-2\gamma)$ and vanishes as $\gamma \to 1/2$. □

Figure 1 illustrates both results: Theorem 3.8 guarantees no non-diagonal state achieves orthogonality, while the distribution of minimum principal angles shows the typical scale of the departure.

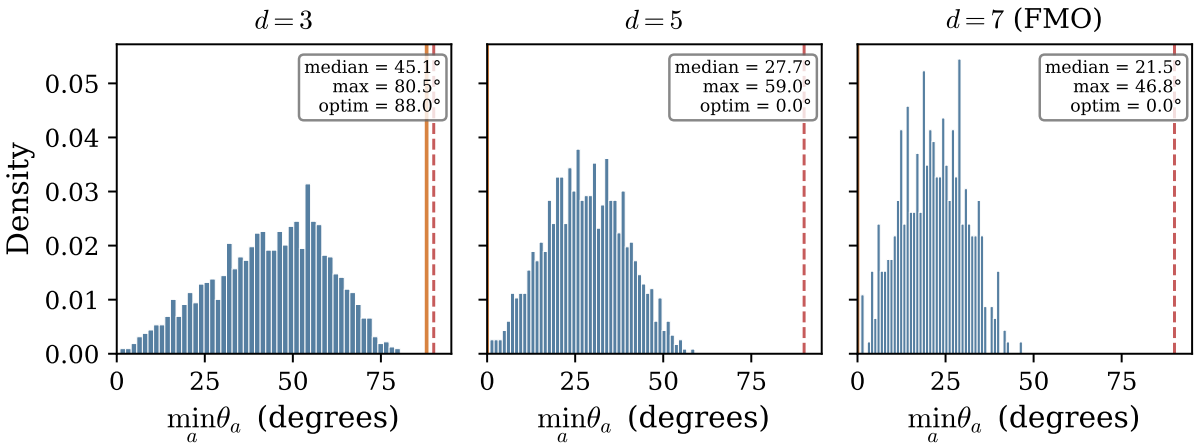

Figure 1: **Numerical illustration of Theorem 3.8.** For each dimension $d \in \{3, 5, 7\}$, random full-rank density matrices are drawn from the Hilbert–Schmidt measure (blue histogram), and the minimum principal angle $\min_a \theta_a$ between the diagonal and off-diagonal tangent subspaces is computed via the Bures metric. The dashed red line marks 90° (perfect orthogonality, achieved only at diagonal states). No non-diagonal state approaches orthogonality; the distribution shifts further from 90° as d increases. The orange line marks the best angle found by adversarial optimization. The $d = 7$ panel corresponds to the FMO Hilbert space dimension.

3.3 Non-classicality scalar

The $d - 1$ principal angles provide a complete geometric characterization of quantum-classical mixing, but a scalar summary is useful for cross-system comparison.

Definition 3.11 (Non-classicality index)**.** For a d-dimensional system with principal angles $\theta_1, \ldots, \theta_{d-1}$ between the diagonal and off-diagonal tangent subspaces, the *non-classicality index* is

$$\chi(\rho) \;=\; \frac{1}{d-1} \sum_{a=1}^{d-1} \cos^2 \theta_a. \tag{18}$$

By construction, $\chi \in [0, 1]$. At diagonal states, all $\theta_a = \pi/2$ (Theorem 3.4), so $\chi = 0$: the system is geometrically classical. Any coherence gives $\chi > 0$ (Theorem 3.8 for $d \geq 3$).

The index χ measures the average squared overlap between classical and quantum tangent directions; it is not a binary classifier but a continuous measure of how far the state's information geometry departs from the classical simplex.

4 Thermodynamic Cost of Decoherence

Proposition 4.1 (Free energy of coherence)**.** *Let ρ be a quantum state with system Hamiltonian H. The free energy change under dephasing is*

$$F(\rho) - F(\Lambda_{\mathrm{dec}}[\rho]) = \mathrm{Tr}\Big[(\rho - \Lambda_{\mathrm{dec}}[\rho])H\Big] + k_B T\, C_r(\rho), \tag{19}$$

where $F(\rho) = \mathrm{Tr}[\rho H] - k_B T S(\rho)$ is the non-equilibrium free energy and $C_r(\rho) = S(\Lambda_{\mathrm{dec}}[\rho]) - S(\rho) \geq 0$ is the relative entropy of coherence [20].

Proof. Direct rearrangement of $F(\rho) - F(\Lambda_{\mathrm{dec}}[\rho])$. Non-negativity of C_r follows from unitality of Λ_{dec} and Schur-concavity of von Neumann entropy. □

When H is diagonal in the dephasing basis, the energy term vanishes and

$$F(\rho) - F(\Lambda_{\mathrm{dec}}[\rho]) = k_B T\, C_r(\rho) \geq 0. \tag{20}$$

The free energy geometrically decomposes along the orthogonal subspaces of Theorem 3.4: the classical component on T_ρ^{diag} and the coherent component on T_ρ^{off}.

For a maximally coherent d-dimensional pure state, $C_r = \ln d$, and the cost per collapsed dimension is $k_B T \ln d/(d^2 - d) \approx k_B T \ln d/d^2$, which vanishes with system size—consistent with the extreme rapidity of macroscopic decoherence [3].

5 Application to Photosynthetic Exciton Transport

5.1 The FMO complex

The Fenna–Matthews–Olson (FMO) complex of green sulfur bacteria (*Chlorobaculum tepidum*) is a trimeric pigment-protein complex that channels excitation energy from the chlorosome antenna to the reaction center. Each monomer contains seven bacteriochlorophyll *a* (BChl *a*) chromophores whose excitonic couplings have been determined spectroscopically [21]. The site-energy Hamiltonian in the site basis (energies in cm^{-1} relative to $12\,210\,\mathrm{cm}^{-1}$) is [21]:

$$H_{\mathrm{FMO}} = \begin{pmatrix} 200 & -87.7 & 5.5 & -5.9 & 6.7 & -13.7 & -9.9 \\ -87.7 & 320 & 30.8 & 8.2 & 0.7 & 11.8 & 4.3 \\ 5.5 & 30.8 & 0 & -53.5 & -2.2 & -9.6 & 6.0 \\ -5.9 & 8.2 & -53.5 & 110 & -70.7 & -17.0 & -63.3 \\ 6.7 & 0.7 & -2.2 & -70.7 & 270 & 81.1 & -1.3 \\ -13.7 & 11.8 & -9.6 & -17.0 & 81.1 & 420 & 39.7 \\ -9.9 & 4.3 & 6.0 & -63.3 & -1.3 & 39.7 & 230 \end{pmatrix}. \tag{21}$$

The dominant energy transfer pathways proceed as $1 \to 2 \to 3$ and $6 \to 5/7 \to 4 \to 3$, with site 3 serving as the exit site coupled to the reaction center [21, 9]. Decoherence due to protein-solvent fluctuations occurs on timescales of $\sim 50\,\mathrm{fs}$ at physiological temperature [9], while excitation transfer completes in ~ 1–$5\,\mathrm{ps}$.

5.2 Dimensional collapse in FMO

For the seven-site FMO complex ($d = 7$):

- Full quantum state space: $d^2 - 1 = 48$ real dimensions
- Classical probability simplex: $d - 1 = 6$ dimensions
- Collapse ratio: $1/(d+1) = 1/8 = 12.5\%$

Decoherence annihilates $d^2 - d = 42$ off-diagonal (coherence) dimensions—seven-eighths of the information-geometric degrees of freedom. The surviving 6-dimensional simplex encodes the classical site populations $\{p_1, \ldots, p_7\}$ (with $\sum_k p_k = 1$), which describe the probability of finding the excitation at each chromophore.

Under Haken–Strobl dephasing [22] at rate γ and trapping at the reaction-centre site 3 at rate Γ_{trap}, the density matrix evolves as

$$\dot{\rho} = -\frac{i}{\hbar}[H_{\text{FMO}}, \rho] + \gamma \sum_{k=1}^{7} \left(|k\rangle\langle k|\rho|k\rangle\langle k| - \frac{1}{2}\{|k\rangle\langle k|, \rho\} \right) - \frac{\Gamma_{\text{trap}}}{2}\{|3\rangle\langle 3|, \rho\}. \tag{22}$$

The first term generates unitary evolution, the second dephases coherences at rate γ, and the third removes population into the reaction-centre sink ($\Gamma_{\text{trap}} = 1\,\text{ps}^{-1}$, following [7]). The transport efficiency is $\eta = \Gamma_{\text{trap}} \int_0^T \langle 3|\rho(t)|3\rangle\, dt$.

Since the trapping term makes the dynamics non-trace-preserving (population leaks irreversibly into the reaction centre), we define the *conditional state* $\hat{\rho}(t) = \rho(t)/\text{Tr}[\rho(t)]$—the state of the exciton conditioned on not yet having been trapped. This renormalized state is a proper density matrix to which the Bures metric and principal angle analysis apply. In the pure-dephasing limit ($H = 0$, $\Gamma_{\text{trap}} = 0$), off-diagonal elements decay as $\rho_{kl} \to e^{-\gamma t}\rho_{kl}$; the full dynamics (22) is a competition between coherent Hamiltonian evolution, incoherent dephasing, and trapping.

5.3 Transport efficiency and ENAQT

ENAQT [7, 6] occurs because pure quantum evolution can trap excitations in interference patterns, while moderate dephasing breaks destructive interference and opens transport channels. Figure 2(a) shows the transport efficiency $\eta(\gamma)$ computed from (22) for both natural entry points: site 1 (from the baseplate) and site 6 (from the CsmA antenna).

Both initial conditions produce a clear ENAQT peak: efficiency rises from low values at $\gamma \to 0$ (quantum localization), reaches a maximum at $\gamma_{\text{opt}} \approx 100$–$150\,\text{cm}^{-1}$, and slowly declines at large γ (classical hopping limit). The optimal dephasing rate falls within the physiological range for FMO at 300 K ($\gamma \sim 50$–$200\,\text{cm}^{-1}$ [9]), consistent with the ENAQT

literature [7].

5.4 Geometry of the quantum-classical split

We now ask: what does the ENAQT regime look like in the information geometry of the quantum-classical transition?

For the FMO complex ($d = 7$), the diagonal and off-diagonal tangent subspaces have dimensions $d - 1 = 6$ and $d^2 - d = 42$ respectively, and their relative orientation is described by $d - 1 = 6$ principal angles $\theta_1 \leq \cdots \leq \theta_6$. We compute these at each dephasing rate γ by evolving the full sink-inclusive model (22) for 5 ps, renormalizing the surviving state $\hat{\rho} = \rho/\mathrm{Tr}[\rho]$, and extracting the Bures-metric cross-Gram matrix between diagonal and off-diagonal basis directions (see Appendix). Figure 2(b) shows the minimum principal angle $\theta_{\min}(\gamma)$ for both site 1 and site 6 initial excitations.

The two initial conditions produce nearly identical angle curves, confirming that the geometric structure of the quantum-classical transition is robust to the choice of entry point. At weak dephasing ($\gamma \lesssim 1\,\mathrm{cm}^{-1}$), the minimum principal angle drops below 50°: the classical population subspace and the quantum coherence subspace are far from orthogonal, meaning that population and coherence parameters are strongly mixed in the Fisher information sense. At moderate dephasing ($\gamma \sim 5$–$20\,\mathrm{cm}^{-1}$), angles are in the range 70°–85°. At the ENAQT optimal dephasing rate ($\gamma_{\mathrm{opt}} \approx 100$–$150\,\mathrm{cm}^{-1}$), all six angles exceed 87° and the state is geometrically close to classical. The dotted vertical lines in Figure 2(b) mark the efficiency-peak γ values from panel (a), showing that optimal transport occurs in a regime where the quantum-classical split is nearly orthogonal. Crucially, both panels are computed from the same sink-inclusive dynamics (22): the efficiency from accumulated trapped population, and the geometry from the renormalized conditional state $\hat{\rho}$.

This quantitative result reveals that ENAQT does not require a regime of strong quantum-classical mixing. The biologically relevant quantum effects arise from the *small residual departure* from classical orthogonality: the system exploits the last few degrees of geometric non-orthogonality before full classical decoupling. The most geometrically

interesting transition—where angles pass through 45°–80°—occurs at dephasing rates ($\gamma \sim 1$–$20\,\mathrm{cm}^{-1}$) well below the physiological range.

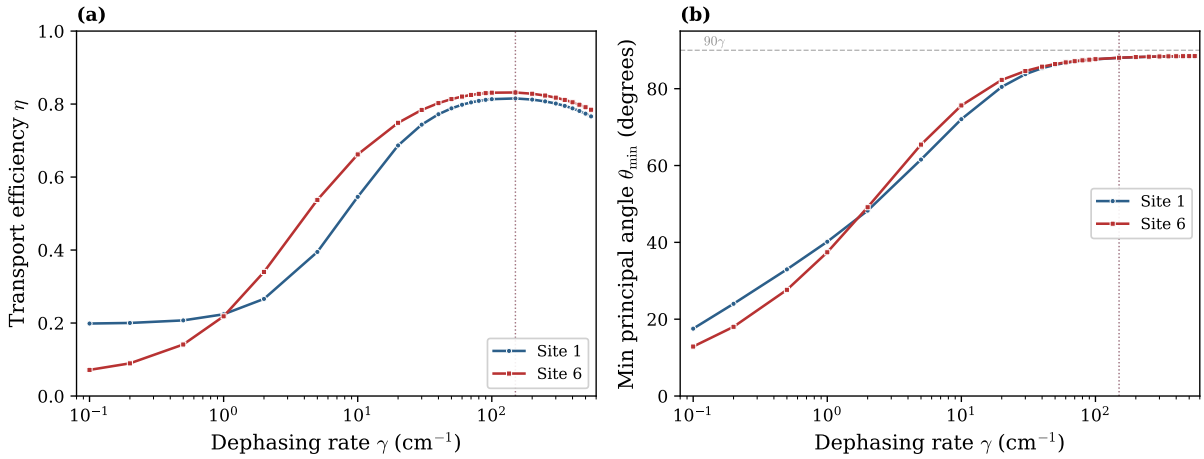

Figure 2: **Transport efficiency and Bures geometry in the FMO complex.** Both panels use the same sink-inclusive dynamics (22) with $\Gamma_{\mathrm{trap}} = 1\,\mathrm{ps}^{-1}$. **(a)** Transport efficiency η as a function of Haken–Strobl dephasing rate γ. Both natural entry points (site 1, site 6) show a clear ENAQT peak at $\gamma_{\mathrm{opt}} \approx 100$–$150\,\mathrm{cm}^{-1}$. **(b)** Minimum Bures principal angle $\theta_{\min}$ between the diagonal and off-diagonal tangent subspaces, computed from the renormalized conditional state $\hat{\rho} = \rho/\mathrm{Tr}[\rho]$ at $t = 5\,\mathrm{ps}$. Dotted vertical lines mark the efficiency-peak γ from panel (a). At the ENAQT optimum, all principal angles exceed 87°: optimal transport occurs in a regime that is geometrically close to classical.

5.5 Comparison to scalar coherence measures

Standard coherence monotones—the ℓ_1-norm $\sum_{k\neq l} |\rho_{kl}|$ and relative entropy C_r—assign single scalars to the quantum-classical transition. They answer "how much coherence?" but not "how is coherence oriented relative to the classical subspace?" The $d-1$ principal angles provide a spectral decomposition of this geometric relationship. Figure 3(a) shows the full six-angle spectrum for FMO: at intermediate dephasing ($\gamma \sim 5$–$20\,\mathrm{cm}^{-1}$), the angles spread substantially, revealing that some subspace directions approach classical orthogonality faster than others. This heterogeneous structure—which directions decohere first and which persist longest—is invisible to any single scalar.

5.6 Robustness to trapping rate

The trapping rate $\Gamma_{\mathrm{trap}} = 1\,\mathrm{ps}^{-1}$ used above follows [7], but the actual rate depends on the FMO-reaction-centre coupling and varies across species. Figure 3(b) shows the minimum

principal angle at the ENAQT-optimal dephasing rate as a function of Γ_{trap} across the range 0.2–5.0 ps^{-1}. For physiologically relevant trapping rates ($\Gamma_{\text{trap}} = 0.5$–$2.0\,\text{ps}^{-1}$), the minimum angle exceeds 86° for both initial conditions. Even at the aggressive rate $\Gamma_{\text{trap}} = 5\,\text{ps}^{-1}$, the angle remains above 81°—still well into the near-classical regime. The geometric conclusion is robust: faster trapping depletes population before dephasing fully completes, reducing the angle toward classical, but the qualitative picture holds.

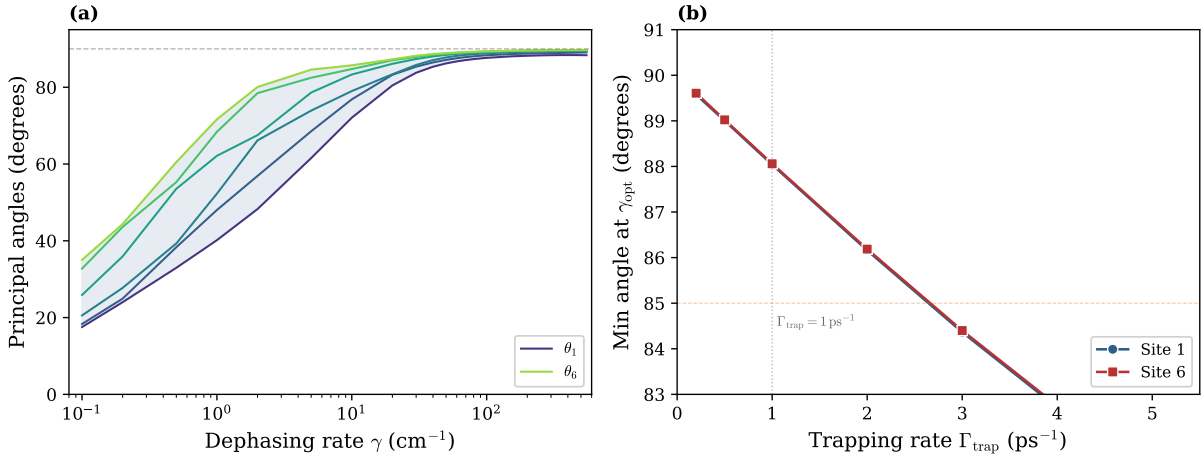

Figure 3: **Angle spectrum and model robustness. (a)** All six Bures principal angles $\theta_1 \leq \cdots \leq \theta_6$ between the diagonal and off-diagonal tangent subspaces as a function of dephasing rate γ (site-1 initial excitation, $\Gamma_{\text{trap}} = 1\,\text{ps}^{-1}$). The spread of angles at intermediate γ reveals heterogeneous geometric mixing across different subspace directions—structure invisible to scalar coherence measures. **(b)** Minimum principal angle at the ENAQT-optimal dephasing rate as a function of trapping rate Γ_{trap}. For physiologically relevant rates (0.5–2.0 ps^{-1}), angles exceed 86°; even at 5.0 ps^{-1} the system remains near-classical ($> 81°$).

5.7 Second complex: PE545

To test whether the geometric signature observed in FMO is specific to that complex or reflects a more general structure, we apply the same analysis to phycoerythrin 545 (PE545), a light-harvesting antenna from the cryptophyte alga *Rhodomonas* CS24 [23]. PE545 differs from FMO in several respects: it contains eight bilin chromophores (rather than seven bacteriochlorophylls), has approximate C_2 pseudo-symmetry, and uses phycoerythrobilin and dihydrobiliverdin pigments with different electronic properties. The excitonic Hamiltonian (Model E of [24]; see Appendix B) has site energies spanning $\sim 1\,600\,\text{cm}^{-1}$—roughly four times the range in FMO—with maximum inter-site couplings

of $92\,\mathrm{cm}^{-1}$.

For PE545, decoherence collapses the full quantum state space from $d^2 - 1 = 63$ dimensions to $d - 1 = 7$ classical dimensions (collapse ratio $1/9 \approx 11\%$). We model the irreversible step as Lindblad trapping at the two lowest-energy chromophores DBV19A and DBV19B (sites 1 and 2), which serve as the terminal energy acceptors in the antenna [24, 23]. The total trapping rate is $\Gamma_{\mathrm{trap}} = 1\,\mathrm{ps}^{-1}$, split equally between the two target sites. Initial excitation is placed at each of the two central-dimer sites PEB50/61C (site 0) and PEB50/61D (site 5), which absorb strongly and initiate energy transfer [25].

Figure 4(a) shows that PE545 exhibits a clear throughput-optimal dephasing regime: transport efficiency peaks at $\gamma_{\mathrm{opt}} \approx 700$–$750\,\mathrm{cm}^{-1}$ for both initial conditions. The optimum occurs at substantially higher dephasing than in FMO ($\gamma_{\mathrm{opt}} \approx 100$–$150\,\mathrm{cm}^{-1}$), consistent with the larger site-energy disorder requiring more noise to bridge energy gaps. Figure 4(b) shows the seven Bures principal angles as a function of dephasing. At γ_{opt}, all seven angles exceed $89°$: the system is geometrically nearly classical at the transport optimum, reproducing the same qualitative signature observed in FMO. The conclusion is robust to the choice of target manifold (including PEB82C as a third target site shifts γ_{opt} by $< 50\,\mathrm{cm}^{-1}$ and changes minimum angles by $< 0.3°$) and to halving the trapping rate.

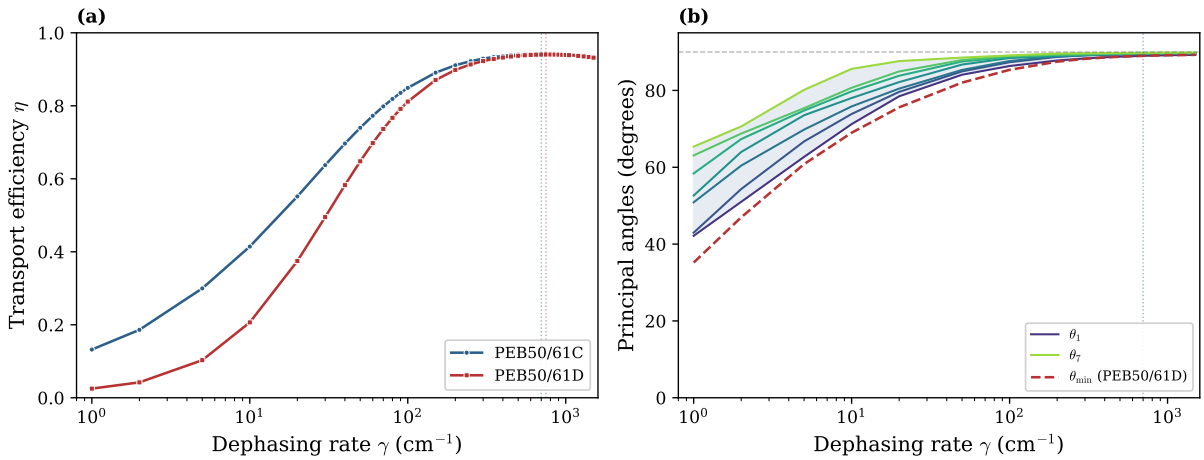

Figure 4: **Transport efficiency and Bures geometry in PE545.** Same analysis as Figure 2, applied to the eight-bilin PE545 complex ($d = 8$, collapse $63 \to 7$). Trapping at the two lowest-energy bilins (DBV19A, DBV19B) with $\Gamma_{\text{trap}} = 1\,\text{ps}^{-1}$. **(a)** Transport efficiency η as a function of dephasing rate γ for both central-dimer initial sites. The throughput-optimal regime peaks at $\gamma_{\text{opt}} \approx 700$–$750\,\text{cm}^{-1}$. **(b)** All seven principal angles for PEB50/61C initial excitation (solid, coloured by angle index), with the minimum principal angle for PEB50/61D overlaid (dashed red). The dotted vertical line marks γ_{opt} from panel (a). At the transport optimum, all angles exceed 89° for both initial conditions—the same near-classical geometric signature as in FMO.

The observation that two structurally distinct photosynthetic complexes—a bacteriochlorophyll trimer antenna (FMO, $d = 7$) and a bilin-based cryptophyte antenna (PE545, $d = 8$)—both exhibit near-classical Bures geometry at their respective transport optima suggests that this geometric property may be a generic feature of environment-assisted transport rather than an accident of FMO's particular Hamiltonian.

5.8 FMO coupling graph

The coupling structure of the FMO Hamiltonian can be represented as a graph where vertices are chromophore sites and edges connect pairs with significant electronic coupling ($|J_{kl}| > 5\,\text{cm}^{-1}$). Figure 5 shows this graph. Four site pairs have couplings below the $5\,\text{cm}^{-1}$ threshold: (2,5), (2,7), (3,5), and (5,7), all involving sites at opposite ends of the two main transport pathways $1 \to 2 \to 3$ and $6 \to 5/7 \to 4 \to 3$.

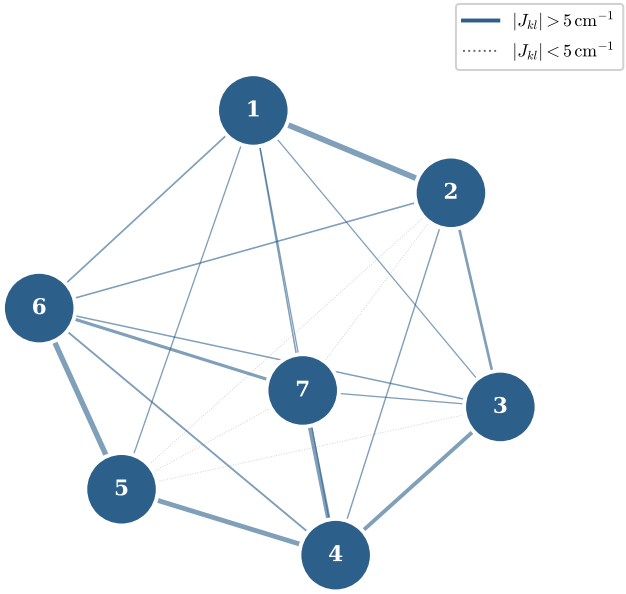

Figure 5: **FMO coupling graph.** Vertices represent the seven BChl sites; edges connect pairs with significant coupling ($|J_{kl}| > 5\,\mathrm{cm}^{-1}$). Edge widths are proportional to coupling strength. Dotted lines show weak couplings ($|J_{kl}| < 5\,\mathrm{cm}^{-1}$). The four weak pairs all involve sites at opposite ends of the two main transport pathways.

6 Application to Neural Oscillations

The principal angle framework developed in Sections 3–5 provides a continuous measure of "how quantum" any system is. Photosynthetic complexes sit near the transition (87°–89° at their respective ENAQT optima). Where do neural oscillator assemblies sit on this spectrum?

6.1 Neural oscillator assemblies

Working memory in primate cortex relies on high-dimensional mixed selectivity: individual neurons respond to conjunctions of task variables, and the population-level representation lives on a high-dimensional manifold [26, 27]. One prominent account attributes this dimensionality to fast gamma-band (30–70 Hz) oscillatory bursts [28, 29, 30], implying that high frequency carries high-dimensional content. The principal angle framework suggests a different interpretation. Lower-frequency oscillations (δ, theta, infraslow) have

smaller $J_{\max}$ and therefore even larger $\gamma_{\text{bio}}/J_{\max}$ than gamma, placing them deeper into the classical regime—meaning they are *more* reliable as information channels, not less. At the same time, low-frequency oscillations coordinate larger cortical volumes [31], entraining more neurons and therefore more ion channels. If each ion channel ($d \sim 4$) sits near the quantum-classical boundary (§6.3), then the effective dimensionality of the coordinated molecular manifold scales with the *number* of entrained sites, not with oscillation frequency. In this view, low frequency implies high dimensionality: not because the oscillation itself is high-dimensional, but because its deep classicality and spatial reach let it reliably coordinate a large manifold of near-quantum molecular systems (*hypothesis*).

A cortical column can be modelled as $d \sim 10$–100 coupled oscillator populations [31]. The question is: where does such an assembly sit on the quantum-classical spectrum measured by the principal angle framework?

6.2 Dimensional collapse in a gamma-band microcircuit

We construct a $d = 10$ Hamiltonian representing ten coupled oscillator populations spanning the gamma band (30–70 Hz; see Appendix D for the full matrix). The coupling topology is a ring with two long-range shortcuts (small-world), matching the recurrent connectivity of cortical microcircuits. For $d = 10$:

- Full quantum state space: $d^2 - 1 = 99$ real dimensions
- Classical probability simplex: $d - 1 = 9$ dimensions
- Collapse ratio: $1/(d + 1) = 1/11 \approx 9\%$

The Lindblad framework is identical to the photosynthetic analysis: Haken–Strobl dephasing at rate γ with trapping at two target sites ($\kappa = 0.5\,\text{ps}^{-1}$ each in scaled units). The crucial property is *scale invariance*: the Lindblad equation under $(H, \gamma, \kappa, t) \to (\alpha H, \alpha\gamma, \alpha\kappa, t/\alpha)$ preserves $\rho(t)$ and therefore all principal angles. Results depend only on the dimensionless ratio $\gamma/J_{\max}$, not absolute energy. We scale the neural Hamiltonian from Hz to cm^{-1} ($\alpha = 50\,\text{cm}^{-1}/\text{Hz}$) so that matrix entries are $O(10^3)$, matching the FMO/PE545 range.

6.3 The detection threshold

At conservative thermal dephasing ($T = 310\,\mathrm{K}$), the biological dephasing rate for neural oscillations is $\gamma_{\mathrm{bio}} = k_B T/\hbar \approx 4 \times 10^{13}\,\mathrm{Hz}$, giving $\gamma_{\mathrm{bio}}/J_{\max} \approx 8 \times 10^{12}$. Since $\theta_{\min} \to 90^\circ$ monotonically as $\gamma/J_{\max} \to \infty$, and simulation at the proxy ratio $\gamma/J_{\max} = 50$ already gives $\theta_{\min} > 89.5^\circ$ (Appendix D), the system is asymptotically indistinguishable from the classical simplex at biological dephasing.

We define a *detection threshold* at $\theta_{\min} = 85^\circ$: below this angle, quantum-classical mixing is at least as strong as FMO at its biological operating point. For the neural system, the detection threshold occurs at $\gamma/J_{\max} \approx 0.09$ (Figure 6b). To reach this regime from biological dephasing, the effective dephasing rate would need to be reduced by ∼14 orders of magnitude from bulk thermal estimates. For comparison, photosynthetic protein baths provide ∼2–3 orders of magnitude of shielding from bulk thermal dephasing [9]—insufficient by ∼11 orders for neural oscillation frequencies.

Crucially, however, "deep classical at oscillation frequencies" does not imply "deep classical at all scales." The molecular machinery that neural oscillations coordinate—ion channels, synaptic receptors, second-messenger cascades—operates at molecular energy scales: backbone and side-chain couplings of 10–50 cm^{-1}, with protein-water dephasing of 100–300 cm^{-1} [32]. This gives $\gamma_{\mathrm{bio}}/J_{\max} \sim O(2\text{–}30)$—the same regime as FMO and PE545, not the deep classical regime of the oscillations they generate (*hypothesis for real channels*). To test this numerically, we include an illustrative $d = 4$ ion-channel selectivity-filter Hamiltonian (Appendix C.1): at $\gamma_{\mathrm{bio}}/J_{\max} = 6.7$, we obtain $\theta_{\min} \approx 88.5^\circ$ and $\chi \approx 2.8 \times 10^{-4}$ (*model-based empirical*), i.e. non-classical but near-classical geometry. The deep classicality of neural oscillations is then a *feature*: the oscillation layer provides a geometrically reliable channel—indistinguishable from the Fisher–Rao simplex—that faithfully transmits molecular-scale events to population-level dynamics without introducing quantum-classical mixing of its own. Tegmark [33] reached the same conclusion about oscillation-frequency scales via decoherence timescales; our contribution is twofold: a continuous geometric bound rather than a binary timescale comparison, and a quantitative non-classicality measure χ that can be carried across scales.

6.4 The geometric spectrum across biological scales

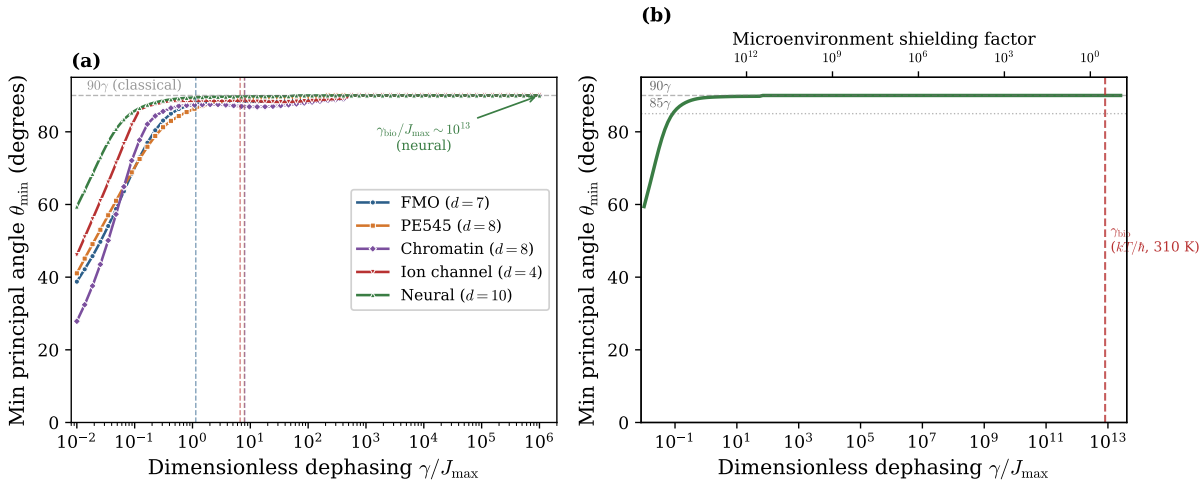

Figure 6: **Cross-scale quantum-classical boundary. (a)** Minimum Bures principal angle $\theta_{\min}$ as a function of dimensionless dephasing $\gamma/J_{\max}$ for FMO ($d = 7$, blue), PE545 ($d = 8$, orange), illustrative ion channel ($d = 4$, red), illustrative chromatin ($d = 8$, purple; Appendix C), and cortical gamma oscillations ($d = 10$, green). Solid lines show Lindblad-simulated values ($\gamma/J_{\max} \leq 500$); beyond this range, angles are set to 90° on asymptotic grounds (the state is numerically indistinguishable from the classical simplex). Dashed vertical lines mark biological dephasing for each molecular system; neural biological dephasing ($\gamma_{\text{bio}}/J_{\max} \sim 10^{13}$) lies far off-scale. All three validated systems (FMO, PE545, neural) exhibit the same transition form; illustrative molecular models test whether the same geometry appears beyond photosynthesis. **(b)** Detection threshold for the neural system (see §6.3).

Figure 6 places validated and illustrative systems on a single axis of dimensionless dephasing $\gamma/J_{\max}$:

- **FMO** ($d = 7$): $\gamma_{\text{bio}}/J_{\max} \approx 1.1$, near the ENAQT optimum. Principal angles 87°–90°. Quantum effects are geometrically exploitable.
- **PE545** ($d = 8$): $\gamma_{\text{bio}}/J_{\max} \approx 7.9$, slightly past the optimum. Angles $> 89°$. Still in the transition zone.
- **Ion channel (illustrative,** $d = 4$**)**: $\gamma_{\text{bio}}/J_{\max} \approx 6.7$. $\theta_{\min} \approx 88.5°$, $\chi \approx 2.8 \times 10^{-4}$. Non-classical ($\chi > 0$) but near-classical geometry.
- **Neural gamma** ($d = 10$): $\gamma_{\text{bio}}/J_{\max} \approx 8 \times 10^{12}$. Angles asymptotically $\to 90°$. Deep classical.

The three validated systems span 12 orders of magnitude in $\gamma_{\text{bio}}/J_{\max}$, yet all exhibit the same transition form: principal angles rise steeply through the $\gamma/J_{\max} \sim O(10^{-1}\text{–}1)$ range, confirming scale invariance of the principal angle framework. The two illustrative molecular models (ion channel and chromatin) sit in the same near-boundary region at reference parameters; chromatin remains sensitivity-limited across weaker couplings (Table 2).

The result suggests a multi-scale architecture that can be formalized using the non-classicality index χ (Definition 3.11). Neural oscillations operate *within* the Fisher–Rao simplex ($\chi_{\text{osc}} \approx 0$), exploiting classical coherence—oscillatory phase relationships, high-dimensional population dynamics—rather than quantum coherence [34, 35]. But the molecular systems they coordinate—ion channels ($d_m \sim 4$), synaptic receptors, protein complexes—operate at energy scales where $\gamma_{\text{bio}}/J_{\max} \sim O(1\text{–}30)$. The illustrative ion-channel model gives $\chi_m \approx 2.8 \times 10^{-4} > 0$ at biological dephasing; real-channel values remain a testable hypothesis.

Definition 6.1 (Coordinated non-classical dimensionality)**.** Let N molecular subsystems of dimension d_m be entrained by a single oscillatory phase. Each subsystem has $d_m - 1$ classical (diagonal) dimensions and $d_m^2 - d_m$ coherence (off-diagonal) dimensions, with non-classicality index χ_m. The *coordinated non-classical dimensionality* is

$$D_{\text{coord}} \;=\; N\Big[(d_m - 1) + \chi_m\,(d_m^2 - d_m)\Big]. \tag{23}$$

When $\chi_m = 0$ (classical), $D_{\text{coord}} = N(d_m - 1)$: only the simplex dimensions contribute. When $\chi_m > 0$, a fraction of the $d_m^2 - d_m$ coherence dimensions remain geometrically accessible, scaling the effective dimensionality beyond the classical count.

Proposition 6.2 (Low frequency, high dimensionality)**.** (Hypothesis-derived.) *If the oscillation layer is deep classical ($\chi_{\text{osc}} \approx 0$) and the molecular layer has $\chi_m > 0$, then:*

(i) *The oscillation layer transmits molecular-scale events to population-level dynamics without introducing quantum-classical mixing of its own (it acts within the Fisher–Rao simplex).*

(ii) *D_{coord} scales linearly with N, the number of entrained molecular sites.*

(iii) *Since low-frequency oscillations entrain larger cortical volumes [31], N increases with decreasing frequency, and D_{coord} increases accordingly.*

The representational dimensionality of the coordinated molecular manifold scales with entrained site count, not oscillation frequency.

Deep classicality is what makes this scaling possible: the oscillation layer is a geometrically reliable channel ($\chi_{\text{osc}} \approx 0$) that faithfully transmits molecular-scale events with high fidelity precisely because it adds no quantum-classical mixing of its own. The principal angle framework converts "is the nervous system quantum?" from a binary question into a scale-dependent geometric answer: deep classical at oscillation frequencies ($\chi_{\text{osc}} \approx 0$), non-classical at molecular scales ($\chi_m > 0$), with classical oscillations providing the reliable high-fidelity channel between the two. Computing χ_m for experimentally determined ion channel Hamiltonians is a natural next step (§7.4).

As an empirical validation, we repeat the analysis with the Potjans–Diesmann canonical cortical microcircuit model ($d = 8$, empirical connectivity from [36]; see Appendix D). Both the synthetic ($d = 10$) and empirical ($d = 8$) neural models give the same qualitative result: all angles $\approx 90°$ at biological dephasing, with the ENAQT-equivalent regime at $\gamma/J_{\max} \sim O(10^{-1}\text{–}1)$.

7 Discussion

7.1 Summary of results

We have shown that decoherence is a dimensional collapse in the information-geometric sense: $d^2 - 1 \to d - 1$ dimensions, with the Bures metric restricting to Fisher–Rao on the surviving simplex. The exact Bures separation angle quantifies the geometry of the classical-quantum split at every full-rank qubit state, and the global orthogonality theorem (Theorem 3.8) establishes that for $d \geq 3$ the diagonal and off-diagonal tangent subspaces are Bures-orthogonal *only* at diagonal states—any coherence breaks orthog-

onality, resolving the higher-dimensional geometry exactly. Applied to three biological systems spanning 12 orders of magnitude in dephasing—FMO ($d = 7$), PE545 ($d = 8$), and a cortical gamma assembly ($d = 10$)—the principal angle framework reveals a continuous geometric spectrum: photosynthetic complexes sit near the quantum-classical boundary (angles 87°–89°), while neural oscillator assemblies are asymptotically indistinguishable from the classical simplex at oscillation-frequency scales (angles $\to 90°$). For an illustrative ion-channel model ($d = 4$), biological dephasing yields $\theta_{\min} \approx 88.5°$ with $\chi_m \approx 2.8 \times 10^{-4}$: non-classical but near-classical geometry. This supports (without proving for real channels) the molecular-scale hypothesis that $\gamma_{\text{bio}}/J_{\max} \sim O(1\text{–}30)$ can maintain $\chi_m > 0$. The deep classicality of oscillations ($\chi_{\text{osc}} \approx 0$) is then a feature: it provides a reliable channel through which N non-classical molecular sites contribute coordinated dimensionality $D_{\text{coord}} = N[(d_m - 1) + \chi_m(d_m^2 - d_m)]$ (Proposition 6.2), scaling with entrained site count rather than oscillation frequency.

7.2 The dimensional cost-benefit interpretation

The results of Sections 5–6 connect the proved thermodynamic results (§4) with the empirical geometric positions to suggest a cost-benefit interpretation of the ENAQT regime.

Proved: Coherences store free energy $\Delta F = k_B T \cdot C_r(\rho)$ (§4). Decoherence destroys $d^2 - 1 - (d - 1) = d(d - 1)$ dimensions, and the cost per collapsed dimension vanishes as $\sim \ln d/d^2$ for large d.

Empirical: FMO and PE545 transport optima occur at intermediate dephasing where $\theta_{\min} \sim 87°\text{–}89°$ (§5). Neural systems are deep classical at oscillation-frequency scales (§6). An illustrative molecular-oscillator model (Appendix C) sits near the boundary at reference parameters ($\theta_{\min} \approx 87°$), though the position is sensitive to coupling strength.

Hypothesis: The ENAQT regime represents a cost-benefit sweet spot—extra dimensional capacity from partial coherence at modest thermodynamic cost. At physiological temperature ($kT \approx 215\,\text{cm}^{-1}$), molecular couplings of 10–50 cm^{-1} place the dimensionless ratio $\gamma_{\text{bio}}/J_{\max} \sim O(1\text{–}20)$, generically near the quantum-classical boundary. This applies

not only to photosynthetic complexes but to any molecular-scale biological system: ion channel selectivity filters ($d \sim 4$, couplings 10–50 cm^{-1}, dephasing 100–300 cm^{-1}) give $\gamma_{\mathrm{bio}}/J_{\max} \sim O(2\text{–}30)$, placing them in the same geometric regime. Whether this is coincidence or reflects evolutionary optimization is testable: systems like FMO that sit closer to $\gamma_{\mathrm{bio}}/J_{\max} \approx 1$ may be further tuned.

This cost-benefit interpretation does not require defining or optimizing an explicit objective function. It is an observation about the geometric position of biological systems relative to the proved thermodynamic constraints, not a claim of optimality.

7.3 Limitations

The qubit separation angle (Theorem 3.5) is exact, and the global orthogonality theorem (Theorem 3.8) is sharp for all $d \geq 3$; the perturbative expansion (Theorem 3.10) quantifies the departure from orthogonality but is limited to small coherence. The Haken–Strobl dephasing model used for FMO is simplified: the real protein-solvent bath has structured spectral density, non-Markovian dynamics, and spatially correlated fluctuations [9]. The principal angles are computed on the renormalized conditional state $\hat{\rho} = \rho/\mathrm{Tr}[\rho]$; this is the natural object for the surviving exciton's geometry, but it means the Bures analysis inherits any artifacts of the Haken–Strobl model's treatment of system-bath correlations. Extending the information-geometric analysis to structured baths and non-Markovian dephasing is an open problem.

7.4 Outlook

Theorem 3.3 establishes that the Fisher–Rao metric on the surviving classical simplex is the restriction of the Bures metric from the quantum state manifold. The photosynthetic and neural applications demonstrate that this geometric continuity holds across 12 orders of magnitude in dephasing: the same framework that quantifies quantum-classical mixing in photosynthesis also characterizes the detection threshold for quantum effects in neural systems. Classical coherence phenomena on longer timescales [34, 35]—dissipative dynamics within the Fisher–Rao simplex—can in principle be analyzed with the same

information-geometric tools developed here, now that the boundary between quantum and classical regimes has been quantified.

A natural next step is applying the framework to experimentally determined molecular Hamiltonians from 2D-IR spectroscopy, where both the coupling topology and dephasing rates are directly measured. The energy-scale argument (Appendix C) predicts that molecular systems with $kT \sim J_{\max}$ will sit near the quantum-classical boundary, but the sensitivity of this position to coupling strength means that system-specific characterization is needed. The principal angle framework provides a testable geometric tool for this purpose.

The most immediate extension is computing χ_m (Definition 3.11) for ion channel selectivity filters, where the energy-scale argument is sharpest. A potassium channel selectivity filter has $d = 4$ binding sites with ion-ion Coulomb and backbone couplings of 10–50 cm^{-1} and protein-water dephasing of 100–300 cm^{-1} [32], giving $\gamma_{\mathrm{bio}}/J_{\max} \sim O(2$–$30)$ and a dimensional collapse from 15 to 3. If $\chi_m > 0$ in this regime—as the energy-scale argument predicts—then Proposition 6.2 gives a concrete D_{coord} for any entrained population of N channels, and the multi-scale picture becomes quantitative: partial non-classicality at the molecular scale, faithfully propagated by deep classical oscillations ($\chi_{\mathrm{osc}} \approx 0$), influencing neural computation through the phase-coordination machinery described in [34].

7.5 Open questions

1. *Non-Markovian geometry*: characterizing the dimensional collapse under structured (non-Markovian) dephasing, where coherence revivals temporarily restore collapsed dimensions.

2. *Multi-chromophore entanglement*: extending beyond single-exciton states to multi-exciton manifolds, where inter-chromophore entanglement adds structure beyond site-basis coherence.

3. *Quantum Darwinism*: the pointer states selected by einselection [3, 37] determine

which $(d-1)$-dimensional submanifold the collapse targets; the collapse magnitude is basis-independent.

4. *Multi-scale neural architecture*: computing χ_m for ion channel selectivity filters using experimentally determined Hamiltonians, evaluating D_{coord} (Definition 6.1) for realistic entrained populations, and connecting molecular-scale non-classicality to oscillation-level dynamics through the phase-coordination framework of [34].

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A Numerical Methods

Random state generation. Full-rank density matrices are drawn from the Hilbert–Schmidt measure: generate a $d \times d$ matrix A with i.i.d. standard complex Gaussian entries, form $\rho = AA^\dagger/\mathrm{Tr}[AA^\dagger]$.

SLD computation. For each tangent direction X, the SLD L_X satisfying $\rho L_X + L_X \rho = 2X$ is computed in the eigenbasis: $(L_X)^{\mathrm{eig}}_{mn} = 2X^{\mathrm{eig}}_{mn}/(\lambda_m + \lambda_n)$.

Principal angles. The $d-1$ principal angles between T_ρ^{diag} and T_ρ^{off} are computed as $\theta_a = \arccos(\sigma_a)$, where $\sigma_1, \ldots, \sigma_{d-1}$ are the singular values of $G_{DD}^{-1/2} G_{DO} G_{OO}^{-1/2}$.

Adversarial search. 100 L-BFGS-B optimizations per dimension maximize $\min_a \theta_a$ over full-rank parameterized states. No optimized state approaches 90°.

FMO dephasing simulation. The Lindblad master equation (22) is integrated numerically using a fourth-order Runge–Kutta scheme with time step $\Delta t = 1.0\,\mathrm{fs}$. The FMO Hamiltonian (21) uses site energies and couplings from Adolphs and Renger [21]. Both transport efficiency and principal angles are computed from the same sink-inclusive dynamics ($\Gamma_{\mathrm{trap}} = 1\,\mathrm{ps}^{-1}$ at site 3). For the efficiency, accumulated trapped population is integrated over $t = 15\,\mathrm{ps}$. For the principal angles, the density matrix is evolved for $t = 5\,\mathrm{ps}$ and the surviving state is renormalized: $\hat{\rho} = \rho/\mathrm{Tr}[\rho]$. Both initial conditions $\rho(0) = |1\rangle\langle 1|$ (site 1) and $\rho(0) = |6\rangle\langle 6|$ (site 6) are used to test robustness. For dephasing rates $\gamma \geq 5\,\mathrm{cm}^{-1}$, the coherence decay time $\gamma^{-1} \lesssim 1\,\mathrm{ps}$ is shorter than the evolution time, ensuring coherences have reached a quasi-stationary balance. At physiological dephasing

rates ($\gamma \geq 50\,\mathrm{cm}^{-1}$), the minimum principal angle varies by less than 1.2° across readout times of 3–8 ps, confirming that the geometric conclusions are not sensitive to the choice of evolution window. The renormalized state is regularized to full rank ($\hat{\rho} \mapsto \hat{\rho} + 10^{-10}\, I/d$) before computing the six Bures principal angles between the diagonal and off-diagonal tangent subspaces.

Software. All computations use NumPy and SciPy. Source code is available in the repository.

B PE545 Hamiltonian and Parameters

The PE545 Hamiltonian uses site energies and couplings from Model E of Novoderezhkin et al. [24]. Site ordering: 0 = PEB50/61C, 1 = DBV19A, 2 = DBV19B, 3 = PEB82C, 4 = PEB158C, 5 = PEB50/61D, 6 = PEB82D, 7 = PEB158D.

$$H_{\mathrm{PE545}} = \begin{pmatrix} 18532 & 1 & -37 & 37 & 23 & 92 & -16 & 12 \\ 1 & 18008 & 4 & -11 & 33 & -39 & -46 & 3 \\ -37 & 4 & 17973 & 45 & 3 & 2 & -11 & 34 \\ 37 & -11 & 45 & 18040 & -7 & -17 & -3 & 6 \\ 23 & 33 & 3 & -7 & 18711 & 18 & 7 & 6 \\ 92 & -39 & 2 & -17 & 18 & 19574 & 40 & 26 \\ -16 & -46 & -11 & -3 & 7 & 40 & 19050 & 7 \\ 12 & 3 & 34 & 6 & 6 & 26 & 7 & 18960 \end{pmatrix} \mathrm{cm}^{-1}. \quad (24)$$

The diagonal entries are site energies; off-diagonal entries are electronic couplings. Sites 1 and 2 (DBV19A, DBV19B) are the lowest-energy bilins forming the "red pool" that serves as the terminal energy acceptor. Initial excitation is placed at sites 0 or 5 (the central PEB50/61 dimer). The Lindblad trapping sink is applied at the target manifold $\{|1\rangle, |2\rangle\}$ with $\kappa_1 = \kappa_2 = 0.5\,\mathrm{ps}^{-1}$ ($\Gamma_{\mathrm{trap}} = 1.0\,\mathrm{ps}^{-1}$ total). All other model parameters (dephasing model, RK4 integrator, regularization) are identical to the FMO analysis (Appendix A).

Table 1 reports the robustness of PE545 results to target-manifold definition and

trapping rate, backing the claims in Section 5.7.

Table 1: **PE545 robustness check.** Transport optimum γ_{opt}, peak efficiency η, and minimum Bures principal angle $\theta_{\min}$ at γ_{opt} (readout at $t = 5\,\text{ps}$) for three model variants. "Default": target sites $\{1, 2\}$, $\Gamma_{\text{trap}} = 1.0\,\text{ps}^{-1}$. "Alt. target": target sites $\{1, 2, 3\}$, same total rate. "Half κ": target sites $\{1, 2\}$, $\Gamma_{\text{trap}} = 0.5\,\text{ps}^{-1}$.

Condition	Initial site	γ_{opt} (cm^{-1})	η	$\theta_{\min}$
Default	PEB50/61C	725	0.943	89.0°
	PEB50/61D	775	0.941	89.1°
Alt. target	PEB50/61C	725	0.950	89.2°
	PEB50/61D	775	0.947	89.2°
Half κ	PEB50/61C	700	0.800	89.5°
	PEB50/61D	800	0.797	89.5°

C Illustrative Molecular-Oscillator Case Study: Chromatin Dynamics

The photosynthetic results of §5 raise a natural question: is the near-classical geometric position at the ENAQT optimum specific to photosynthetic complexes, or does it reflect a generic feature of molecular biology at physiological temperature? We address this with an illustrative molecular-oscillator case study.

The energy-scale argument. At physiological temperature ($T = 310\,\text{K}$), $kT \approx 215\,\text{cm}^{-1}$. Protein conformational couplings fall in the range 5–50 cm^{-1} [38], and DNA backbone couplings are 10–50 cm^{-1} from THz/IR spectroscopy. This places $\gamma_{\text{bio}}/J_{\max} \sim$ 4–40 for generic molecular-scale biological systems—near the quantum-classical boundary. Whether specific systems exploit this position is a testable question.

The nucleosome-remodeler model. We construct a $d = 8$ illustrative model. The model is physically motivated but not a fit to a specific experimentally determined complex. Eight coupled sites represent: DNA entry/exit loci (sites 0, 1), histone tail contacts

(sites 2, 3), remodeler ATPase contacts (sites 4, 5), and linker DNA modes (sites 6, 7).

Sites. 0 = DNA entry, 1 = DNA exit, 2 = histone tail 1, 3 = histone tail 2, 4 = ATPase 1, 5 = ATPase 2, 6 = linker DNA 1, 7 = linker DNA 2.

Site energies ($\mathbf{cm^{-1}}$). $\{20, 35, 50, 65, 40, 55, 30, 45\}$ (THz protein mode range).

Coupling topology. Open chain with nearest-neighbour coupling $J_{\mathrm{nn}} = 25\,\mathrm{cm}^{-1}$ plus a cross-link $J_{0,7} = 10\,\mathrm{cm}^{-1}$. $J_{\max} = 25\,\mathrm{cm}^{-1}$.

Physical justification.

- Protein conformational coupling: 5–50 cm^{-1} (allosteric signaling literature [38])
- DNA backbone coupling: 10–50 cm^{-1} (THz/IR spectroscopy)
- Protein dephasing in water: 100–300 cm^{-1} (2D-IR, Fleming/Hamm/Tokmakoff groups [32])

Biological dephasing. $\gamma_{\mathrm{bio}} = 200\,\mathrm{cm}^{-1}$ (mid-range of 2D-IR measured protein aqueous dephasing at 310 K). $\gamma_{\mathrm{bio}}/J_{\max} = 8.0$.

Results at reference parameters. At reference parameters ($J_{\max} = 25\,\mathrm{cm}^{-1}$, $\gamma_{\mathrm{bio}} = 200\,\mathrm{cm}^{-1}$), the minimum principal angle is $\theta_{\min} \approx 87°$: the system is geometrically near-classical, consistent with the energy-scale argument. For $d = 8$, decoherence collapses $d^2 - 1 = 63$ quantum dimensions to $d - 1 = 7$ classical dimensions.

Sensitivity sweep. Table 2 reports $\theta_{\min}$ across physically reasonable parameter ranges. At weak coupling ($J_{\max} \leq 15\,\mathrm{cm}^{-1}$), several values drop below 85°, indicating that the geometric position is sensitive to coupling strength. At moderate-to-strong coupling ($J_{\max} \geq 20\,\mathrm{cm}^{-1}$), all values exceed 86°. The sensitivity to coupling strength means that the "molecular biology generically near the boundary" claim requires system-specific validation with experimentally determined Hamiltonians rather than illustrative models.

Table 2: **Chromatin sensitivity sweep.** Minimum Bures principal angle $\theta_{\min}$ (degrees) for each $(\gamma_{\text{bio}}, J_{\max})$ combination. Bold values indicate $\theta_{\min} < 85°$. Hamiltonian couplings are scaled proportionally with $J_{\max}$.

γ_{bio} (cm^{-1})	J=10	J=15	J=20	J=25	J=30	J=40	J=50
100	**82.5**	**84.7**	86.3	87.4	88.0	88.5	88.8
150	**83.2**	**84.8**	86.1	87.1	87.8	88.5	88.8
200	**83.7**	85.0	86.1	87.0	87.7	88.5	88.8
250	**84.2**	85.1	86.1	86.9	87.5	88.4	88.8
300	**84.6**	85.3	86.2	86.9	87.5	88.3	88.8

C.1 Ion Channel Selectivity Filter (Illustrative)

To make the molecular-scale neural hypothesis quantitative, we add an illustrative $d = 4$ ion-channel selectivity-filter model (KcsA-like). The four sites represent binding pockets $S1$–$S4$ in a linear chain with nearest-neighbour coupling $J_{\text{nn}} = 30\,\text{cm}^{-1}$:

$$H_{\text{ion}} = \begin{pmatrix} 30 & 30 & 0 & 0 \\ 30 & 45 & 30 & 0 \\ 0 & 30 & 45 & 30 \\ 0 & 0 & 30 & 30 \end{pmatrix} \text{cm}^{-1}. \tag{25}$$

This gives $J_{\max} = 30\,\text{cm}^{-1}$, and with $\gamma_{\text{bio}} = 200\,\text{cm}^{-1}$ the biological ratio is $\gamma_{\text{bio}}/J_{\max} = 6.7$. Using the same Lindblad pipeline as the main text, we obtain

$$\theta_{\min}^{\text{bio}} \approx 88.55°, \qquad \chi_{\text{ion}}^{\text{bio}} \approx 2.8 \times 10^{-4}.$$

Thus the model is non-classical ($\chi > 0$) but close to the classical simplex. This is a model-based anchor for the molecular-scale hypothesis, not a claim that real channels are fully characterized.

D Neural Microcircuit Parameters

Primary model: gamma-band cortical microcircuit ($d = 10$). Ten coupled oscillator populations spanning the gamma band (30–70 Hz), modelling a cortical column. Site labels: L4in (layer-4 input), L4a, L4b, L2/3a, L2/3b, L2/3c, L5a, L5b, L6a, L6b. Coupling topology: ring with nearest-neighbour coupling ($J = 5\,\mathrm{Hz}$) plus two long-range shortcuts (sites $1 \leftrightarrow 6$ and $3 \leftrightarrow 8$, $J = 2\,\mathrm{Hz}$), producing small-world connectivity consistent with cortical column architecture [31].

Site energies (Hz): $\{30, 33, 36, 40, 44, 48, 53, 58, 64, 70\}$.

The Hamiltonian is scaled to cm^{-1} by $\alpha = 50\,\mathrm{cm}^{-1}/\mathrm{Hz}$ for numerical compatibility with the Lindblad integrator. By scale invariance of the Lindblad equation (Section 6.2), all principal angles depend only on the dimensionless ratio $\gamma/J_{\max}$.

Physical parameters:

- Site energy spread: $\Delta E \approx 40\,\mathrm{Hz}$
- Maximum coupling: $J_{\max} = 5\,\mathrm{Hz}$ (nearest-neighbour)
- Biological dephasing (thermal): $\gamma_{\mathrm{bio}} = k_B T/\hbar \approx 4 \times 10^{13}\,\mathrm{Hz}$ at $310\,\mathrm{K}$
- Dimensionless ratio: $\gamma_{\mathrm{bio}}/J_{\max} \approx 8 \times 10^{12}$

Initial excitation: site 0 (L4 input), analogous to thalamic drive. Target manifold: sites 8, 9 (lowest-frequency populations—energy flows downhill in frequency). Trapping rate: $\kappa = 0.5\,\mathrm{ps}^{-1}$ per target site (in scaled units).

Empirical validation: Potjans–Diesmann cortical column ($d = 8$). The canonical cortical microcircuit model of Potjans and Diesmann [36] provides a published $d = 8$ connectivity matrix: four layers $\times$ {excitatory, inhibitory} = L2/3e, L2/3i, L4e, L4i, L5e, L5i, L6e, L6i. Connection probabilities and relative synaptic weights are from Table 5 of [36]. We symmetrize the connection probability matrix via $J_{ij}^{\mathrm{eff}} = \sqrt{C_{ij}\,C_{ji}}$ and scale to the cm^{-1} range. Initial excitation at L4e (thalamic input); target manifold L5e, L6e (cortical output layers).

Table 3: **Cross-scale comparison.** Optimal dimensionless dephasing $\gamma_{\text{opt}}/J_{\max}$, minimum principal angle at the transport optimum $\theta_{\min}^{\text{opt}}$, and minimum principal angle at biological dephasing $\theta_{\min}^{\text{bio}}$. FMO and PE545 results are *empirical* (validated Hamiltonians); chromatin is *empirical* (illustrative model, Appendix C); neural results are *empirical* (synthetic + empirical models). Neural $\theta_{\min}^{\text{bio}}$ values are proxy-simulated at $\gamma/J_{\max} = 50$; at biological dephasing ($\gamma_{\text{bio}}/J_{\max} \sim 10^{13}$), angles are asymptotically $\to 90°$.

System	d	$d^2-1 \to d-1$	$\gamma_{\text{opt}}/J_{\max}$	$\theta_{\min}^{\text{opt}}$	$\theta_{\min}^{\text{bio}}$
FMO	7	$48 \to 6$	$\sim$1–2	87°–89°	87°–88°
PE545	8	$63 \to 7$	~ 8	89°	89°
Chromatin (illustrative)	8	$63 \to 7$	~ 0.6	87°	87°
Neural (synthetic)	10	$99 \to 9$	~ 4	89.7°	$> 89.5°^{\dagger}$
Neural (PD empirical)	8	$63 \to 7$	~ 1	89.4°	$> 89.5°^{\dagger}$

†Proxy simulation at $\gamma/J_{\max} = 50$; asymptotically $\to 90°$ at biological $\gamma_{\text{bio}}/J_{\max} \sim 10^{13}$.

References for neural parameters: gamma oscillation mechanisms and cortical column architecture [31]; gamma/beta bursts in working memory [28]; working memory 2.0 [29]; high-dimensional mixed selectivity [27, 26]; cytoelectric field dynamics [30]; canonical cortical microcircuit [36]; decoherence timescale estimates for neural systems [33].